

# SIMATS ENGINEERING ASSESSMENT TOOLS

**COURSE NAME:** Theory of Computation **COURSE CODE:** CSA1304

## **COURSE OUTCOME COVERED**

**CO4: Construct Context-Free Grammars, generate derivations and parse trees to demonstrate the syntactic structure of strings. (BL : 3)**

*Assessment Tool Weightage: 10%*

**QUESTIONS: 10 Total Marks:100 Data Interpretation**

### **Code of Conduct:**

*I **Girija B /192311044** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

*I understand that any violation of academic integrity rules will result in disciplinary*

*action.*

**Signature of the student: GIRIJA B**

# SIMATS ENGINEERING ASSESSMENT TOOLS

## Given Data (Grammar)

Consider the CFG:

$S \rightarrow aSb \mid ab$

## Q1. Language Identification

Based on the given grammar, which of the following strings belong to the language? A) ab

B) aabb

C) aaabbb

D) abb

## ANSWER:

The grammar generates strings with equal numbers of as and bs, with all as before the bs:

- $S \rightarrow ab \rightarrow ab$
- $S \rightarrow aSb \rightarrow a(ab)b \rightarrow aabb$
- $S \rightarrow aSb \rightarrow aaSbb \rightarrow aa(ab)bb \rightarrow aaabbb$

Therefore, the correct answers are:

- ☒ A) ab
- ☒ B) aabb
- ☒ C) aaabbb

## Q2. Pattern Recognition

From the grammar, identify the correct pattern:

A)  $a^n b^n$

B)  $(ab)^n$

C)  $a^n b^m$

D)  $(a+b)^*$

Answer: A)  $a^n b^n$

## Explanation:

The grammar  $S \rightarrow aSb \mid ab$  generates strings with **equal numbers of as followed by equal numbers of bs**.

Examples: ab, aabb, aaabbb

Therefore, the pattern is  $a^n b^n$ .

## Q3. Derivation Analysis

Which of the following is a valid derivation for aabb?

A)  $S \rightarrow aSb \rightarrow aaSbb \rightarrow aabb$

B)  $S \rightarrow ab \rightarrow aabb$

C)  $S \rightarrow aSb \rightarrow ab$

D)  $S \rightarrow aaSbb \rightarrow aabb$

**Answer: A)  $S \rightarrow aSb \rightarrow aaSbb \rightarrow aabb$**

**Short Explanation:**

Starting from S:

$S \rightarrow aSb \rightarrow aaSbb \rightarrow aa(ab)bb \rightarrow aabb$

Therefore, **A is the valid derivation.**

#### **Q4. Missing Step Identification**

**Fill in the missing step:**

$S \rightarrow aSb \rightarrow aaSbb \rightarrow \underline{\hspace{2cm}} \rightarrow aabb$

A) ab

B) S

C)  $\epsilon$

D) aSb

**Answer: A) ab**

**Explanation:**

Using  $S \rightarrow ab$ :

$S \rightarrow aSb \rightarrow aaSbb \rightarrow aa(ab)bb \rightarrow aabb$

So the missing step is **ab**.

#### **Q5. Parse Tree Interpretation**

**How many levels are there in the parse tree for string aaabbb?**

A) 2

B) 3

C) 4

D) 5

**Answer: D) 5**

**Explanation:**

For aaabbb:

$S \rightarrow aSb \rightarrow aaSbb \rightarrow aaSbbb \rightarrow aaabbb$

The parse tree has **5 levels**.

## **SIMATS ENGINEERING ASSESSMENT TOOLS**

**Consider:**

$S \rightarrow SS \mid a$

#### **Q6. Ambiguity Check**

**This grammar is:**

A) Unambiguous

B) Ambiguous

C) Regular

D) Deterministic

**Answer: B) Ambiguous**

**Explanation:**

The grammar  $S \rightarrow SS \mid a$  is **ambiguous** because some strings can have **more than one parse tree**.

**Q7. String Generation**

**Which string can be generated?**

- A) a
- B) aa
- C) aaa
- D) All of the above

**Answer: D) All of the above**

**Explanation:**

Using  $S \rightarrow SS \mid a$ , we can generate:

- $a \rightarrow$  directly from  $S \rightarrow a$
- $aa \rightarrow S \rightarrow SS \rightarrow aa$
- $aaa \rightarrow S \rightarrow SS \rightarrow aSS \rightarrow aaa$

So, **all three strings can be generated.**

**Q8. Interpretation of Ambiguity**

**Ambiguity means:**

- A) One string  $\rightarrow$  one parse tree
- B) One string  $\rightarrow$  multiple parse trees
- C) No derivation
- D) Infinite symbols

**Answer: B) One string  $\rightarrow$  multiple parse trees**

**Explanation:**

A grammar is **ambiguous** when a single string can have **more than one valid parse tree** or derivation.

**Final Answer: B) One string  $\rightarrow$  multiple parse trees.**

**Given Data (Grammar 1)**

$S \rightarrow 0S1 \mid 01$

**Q9. Derivation Interpretation**

**Which of the following derivations correctly generates the string**

**0011? A)  $S \rightarrow 0S1 \rightarrow 00S11 \rightarrow 0011$**

**B)  $S \rightarrow 01 \rightarrow 0011$**

**C)  $S \rightarrow 0S1 \rightarrow 01$**

**D)  $S \rightarrow 00S11 \rightarrow 0011$**

**Answer: A)  $S \rightarrow 0S1 \rightarrow 00S11 \rightarrow 0011$**

**Explanation:**

Using  $S \rightarrow 01$  in the final step:

$S \rightarrow 0S1 \rightarrow 0(01)1 \rightarrow 0011$

**Final Answer: A.**

**Given Data (Grammar 2)** $S \rightarrow aS \mid Sb \mid ab$ **Q10. Structural Analysis**

Which of the following strings can be generated by the grammar? A) ab

B) aab

C) abb

D) All of the above

**Answer: D) All of the above**

**Explanation:**

The grammar  $S \rightarrow aS \mid Sb \mid ab$  generates strings with one or more as followed by one or more bs.

- ab ☒
- aab ☒  $\rightarrow S \rightarrow aS \rightarrow a(ab)$
- abb ☒  $\rightarrow S \rightarrow Sb \rightarrow (ab)b$

**Final Answer: D) All of the above.**

## SIMATS ENGINEERING ASSESSMENT TOOLS

**RUBRICS FOR CALCULATION**

| Criteria                        | Weightage | Level 4 – Exemplary (5 Marks)                                | Level 3 – Proficient (4 Marks)       | Level 2 – Developing (2–3 Marks)             | Level 1 – Beginning (0–1 Mark)  |
|---------------------------------|-----------|--------------------------------------------------------------|--------------------------------------|----------------------------------------------|---------------------------------|
| <b>Conceptual Understanding</b> | <b>30</b> | Demonstrates clear and complete understanding of the concept | Good understanding with minor gaps   | Basic understanding with some misconceptions | Poor or incorrect understanding |
| <b>Accuracy of Answer</b>       | <b>25</b> | Completely correct answer with no errors                     | Mostly correct with minor errors     | Partially correct with noticeable errors     | Incorrect or irrelevant answer  |
| <b>Application of Knowledge</b> | <b>20</b> | Correctly applies concepts to solve the question/problem     | Applies concepts with minor mistakes | Limited or partially correct application     | No or incorrect application     |

|                                   |           |                                           |                                        |                                       |                                |
|-----------------------------------|-----------|-------------------------------------------|----------------------------------------|---------------------------------------|--------------------------------|
| <b>Completeness of Response</b>   | <b>15</b> | Covers all required parts of the question | Covers most parts with minor omissions | Covers only some parts                | Incomplete or missing response |
| <b>Clarity &amp; Presentation</b> | <b>10</b> | Clear, concise, and well organized answer | Understandable with minor issues       | Somewhat unclear or poorly structured | Difficult to understand        |